Examiner only

2. (a) Crude oil is formed from simple marine organisms.

Tick (✓) the box next to the length of time it takes for the organisms to turn into crude oil. [1]

hundreds of years ☐

thousands of years ☐

millions of years ☐

(b) Crude oil can be separated into simpler mixtures called fractions.

Tick (✓) the box next to the method used to separate crude oil into fractions. [1]

fractional distillation ☐

filtration ☐

cracking ☐

polymerisation ☐

Examiner only

(c) The table shows properties of some fractions that are obtained from crude oil.

Fraction	Size of molecules (chain length)	Viscosity	Ease of ignition	Amount of smoke formed
petrol	$C_5 - C_{10}$	very runny	very easy	no smoke
naphtha	$C_8 - C_{12}$	fairly runny	quite easy	little smoke
kerosene	$C_{10} - C_{16}$	thick	quite hard	quite a lot of smoke
diesel oil	$C_{14} - C_{20}$	very thick	very hard	very smoky

Use only the information from the table to answer the following questions.

(i) Name the fraction which is the easiest to pour. .... [1]

(ii) Name the fraction which is the hardest to burn. .... [1]

(iii) Name the fraction with the smallest **range** of chain lengths. [1]

....

(iv) Name the fraction which burns with the cleanest flame. Give the reason for your choice. [2]

Fraction ....

Reason ....

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